

SQL JOIN Practice – Tables and 10 Questions

Questions

1. Write an INNER JOIN query to display every employee's ID, name, department name and department location. Employees without a department should not appear.

```
SELECT e.emp_id,e.emp_name,d.dept_name,d.location FROM employees e  
INNER JOIN departments d ON e.dept_id = d.dept_id;
```

2. Write a LEFT JOIN query to display all employees along with their department names. Employees who are not assigned to a department must also be displayed.

```
SELECT e.emp_id,e.emp_name,d.dept_name FROM employees e  
LEFT JOIN departments d ON e.dept_id = d.dept_id;
```

3. Write a LEFT JOIN query to find departments that do not currently have any employees

```
SELECT d.dept_id,d.dept_name FROM departments d  
LEFT JOIN employees e ON d.dept_id = e.dept_id WHERE e.emp_id IS NULL;
```

4. Write a query using multiple joins to display the employee name, project name, department name and number of hours worked.

```
SELECT e.emp_name,p.project_name,d.dept_name,ep.hours FROM employees e  
INNER JOIN employee_project ep ON e.emp_id = ep.emp_id  
INNER JOIN projects p ON ep.project_id = p.project_id  
INNER JOIN departments d ON p.dept_id = d.dept_id;
```

5. Write a query to display all projects and the employees assigned to each project. Projects without assigned employees must also appear.

```
SELECT p.project_id,p.project_name,e.emp_name FROM projects p
LEFT JOIN employee_project ep ON p.project_id = ep.project_id
LEFT JOIN employees e ON ep.emp_id = e.emp_id;
```

6. Using a self-join on the employees table, display each employee's name and their manager's name. Employees without managers must also appear

```
SELECT e.emp_name AS employee_name,m.emp_name AS manager_name FROM
employees e
LEFT JOIN employees m ON e.manager_id = m.emp_id;
```

7. Write a query to find employees who are working on more than one project. Display the employee ID, employee name and total number of projects.

```
SELECT e.emp_id,e.emp_name,COUNT(ep.project_id) AS total_projects FROM employees e
INNER JOIN employee_project ep ON e.emp_id = ep.emp_id
GROUP BY e.emp_id,e.emp_name HAVING COUNT(ep.project_id) > 1;
```

8. Write a query to display each department's name, number of employees and average employee salary. Departments without employees must show an employee count of 0.

```
SELECT d.dept_name,COUNT(e.emp_id) AS employee_count,AVG(e.salary) AS
average_salary FROM departments d
LEFT JOIN employees e ON d.dept_id = e.dept_id GROUP BY d.dept_id,d.dept_name;
```

9. Write a query to find employees who work on projects belonging to a department different from their own department. Display the employee name, employee department, project name and project department.

```
SELECT e.emp_name,ed.dept_name AS  
employee_department,p.project_name,pd.dept_name AS project_department FROM  
employees e  
  
INNER JOIN employee_project ep ON e.emp_id = ep.emp_id  
  
INNER JOIN projects p ON ep.project_id = p.project_id  
  
LEFT JOIN departments ed ON e.dept_id = ed.dept_id  
  
INNER JOIN departments pd ON p.dept_id = pd.dept_id WHERE e.dept_id <> p.dept_id;
```

10. Write a query to find projects that have no employees assigned. Display the project ID, project name, department name and budget.

```
SELECT p.project_id,p.project_name,d.dept_name,p.budget FROM projects p  
  
INNER JOIN departments d ON p.dept_id = d.dept_id  
  
LEFT JOIN employee_project ep ON p.project_id = ep.project_id WHERE ep.emp_id IS NULL;
```